# A Restricted Jacobi Identity (Elementary Checked Boundary Case) for the Field-Level Tethered Bracket on the Divergence-Free Shear Class

Companion paper I (machine-checked, finite-Fourier model)

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**Abstract**

In a finite-Fourier model of vector fields on $\mathbb{T}^3$ we consider the vector-valued tethered bracket $\{a,b\} = [a,b] + T(a,b)$, where $[\cdot,\cdot]$ is the vector-field Lie bracket and $T(a,b) = -(\kappa w)\langle \Pi_u a, \Pi_u b\rangle u$. For every divergence-free axis $u$ pointing along $e_x$ with $\langle u,u\rangle \neq 0$, in particular $u = (\sin y, 0, 0)$, we prove that the Jacobiator vanishes whenever all three arguments are divergence-free and point along $e_x$. On this class the classical Lie bracket vanishes. The tether term $T$ need not vanish, but it takes values in $\mathbb{C}u$, which lies in the class, and all nested terms vanish. The result is formalized as `shear_jacobi_holds_of_pr_ne_zero` (and, for $u_Y$, `shearY_jacobi_holds`) in Lean 4/Mathlib. We also prove in Lean that the hypotheses are inhabited with a *nonzero* bracket inside the class, and that the condition $\langle u,u\rangle \neq 0$ cannot be dropped: for the isotropic axis $e^{iy}e_x$ the Jacobiator on $(2\cos y, 0, 0)$ is $6(\kappa w)^2 e^{iy}e_x$. It is an elementary, machine-checked boundary case, not a new Jacobi theorem. Companion paper II shows that the XOnly hypothesis cannot be dropped: an explicit divergence-free diagonal triple fails. We make no claim about the geometric reduced coadjoint orbit.

## 1 The finite-Fourier field model (exact Lean definitions)

All definitions are in `FTVerified/May/FieldJacobi.lean` (namespace `Frohman.Tether.FieldJacobi`), unchanged since commit `26f16da`. For $k \in \mathbb{Z}^3$ write $k_{\mathbb{C}} \in \mathbb{C}^3$ for its image (`kC`, `BiotSavart.lean:46`); $v \cdot w = \sum_j v_j w_j$ is the *bilinear* (non-Hermitian) dot product on $\mathbb{C}^3$. Components are indexed $0,1,2$ $(= x,y,z)$.

**Definition 1** (`Fld`, line 48)**.** `Fld` is the space of finitely supported maps $a : \mathbb{Z}^3 \to \mathbb{C}^3$, $k \mapsto \hat{a}_k$; we read $a$ as the trigonometric polynomial $\sum_k \hat{a}_k e^{ik\cdot x}$ on $\mathbb{T}^3$. No reality condition is imposed; scalars are complex.

**Definition 2** (`bvec`, `br`, lines 51–56)**.** For modes $p,q$ and $v,w \in \mathbb{C}^3$, $\mathrm{bvec}(p,q,v,w) = i\big((v\cdot q)\,w - (w\cdot p)\,v\big)$, and $[a,b] = \sum_{p,q} \mathrm{bvec}(p,q,\hat{a}_p,\hat{b}_q)\,\delta_{p+q}$, the Fourier form of $(a\cdot\nabla)b - (b\cdot\nabla)a$.

**Definition 3** (`pr`, line 59)**.** $\langle a,b\rangle = \sum_p \hat{a}_p \cdot \hat{b}_{-p}$ (complex-bilinear $L^2$ pairing $(2\pi)^{-3}\int a\cdot b$).

**Definition 4** (`piF`, `tetF`, `brT`, `jacT`; lines 108, 111, 362, 364)**.** For $u, x \in \mathsf{Fld}$ and $\kappa, w \in \mathbb{C}$: $\Pi_u x = x - \frac{\langle u,x\rangle}{\langle u,u\rangle} u$ (Lean division, so $\Pi_u x = x$ if $\langle u, u\rangle = 0$); $T(a,b) = -(\kappa w)\langle \Pi_u a, \Pi_u b\rangle\, u$; $\{a,b\} = [a,b] + T(a,b)$; $J(a,b,c) = \{\{a,b\},c\} + \{\{b,c\},a\} + \{\{c,a\},b\}$. Here $\kappa, w$ are *arbitrary complex parameters*: in this file $w$ is not tied to any field (the docstring of `Basic.lean` reads $w$ as $\int |\omega|^2$, but nothing in `FieldJacobi.lean` enforces that).

**Definition 5** (`mixedF`, `mixedSumF`; lines 114, 116)**.** $M(a,b,c) = [T(a,b),c] + T([a,b],c)$ and $S(a,b,c) = M(a,b,c) + M(b,c,a) + M(c,a,b)$.

**Definition 6** (`DivergenceFree`, line 121; `XOnly`, line 183)**.** $a$ is *divergence-free* if $k_{\mathbb{C}} \cdot \hat{a}_k = 0$ for all $k$; $a$ is *XOnly* if $(\hat{a}_k)_1 = (\hat{a}_k)_2 = 0$ for all $k$.

**Definition 7** (explicit fields, lines 224–234)**.** $k_x = (1,0,0)$, $k_y = (0,1,0)$. $u_Y = \delta_{k_y}(-\frac{\mathrm{i}}{2},0,0) + \delta_{-k_y}(\frac{\mathrm{i}}{2},0,0)$, i.e. $(\sin y, 0, 0)$; $a_Z = \delta_{k_x}(0,0,-\frac{\mathrm{i}}{2}) + \delta_{-k_x}(0,0,\frac{\mathrm{i}}{2})$, i.e. $(0,0,\sin x)$.

**Lemma 8** (Jacobiator reduction, `jacT_eq_mixedSum`, line 384)**.** *If $\langle u,u\rangle \neq 0$ then $J(a,b,c) = S(a,b,c)$ for all $a,b,c$ and all $\kappa, w$.*

*Proof.* $T$ is additive in its first slot (`tetF_add_left`) and $[\cdot,\cdot]$ is additive in each slot, so $\{\{a,b\},c\} = [[a,b],c] + M(a,b,c) + T(T(a,b),c)$. Since $T(a,b) = s\,u$ and $\Pi_u(su) = su - \frac{s\langle u,u\rangle}{\langle u,u\rangle}u = 0$ when $\langle u,u\rangle \neq 0$ (`piF_smul_self`), $T(T(a,b),c) = 0$ (`tetF_tetF`). Summing cyclically, $J = \mathrm{jacF}(a,b,c) + S(a,b,c)$, where jacF is the Jacobiator of $[\cdot,\cdot]$. Finally $\mathrm{jacF} \equiv 0$ (`jacF_eq_zero`, line 164): by trilinearity it suffices to check single modes, where all three terms sit at mode $p+q+r$ and the identity is a polynomial identity in the coordinates of $v, w, z, p, q, r$ (closed by `ring`); it is the Fourier form of Jacobi for the Lie bracket of vector fields. □

**Lemma 9** (`pr_uY_uY`, line 396)**.** $\langle u_Y, u_Y\rangle = \frac{1}{2}$*; in particular* $\neq 0$.

*Proof.* $\langle u_Y, u_Y\rangle = (-\frac{\mathrm{i}}{2})(\frac{\mathrm{i}}{2}) + (\frac{\mathrm{i}}{2})(-\frac{\mathrm{i}}{2}) = \frac{1}{4} + \frac{1}{4}$. □

## 2 Main theorem

**Lemma 10** (`br_eq_zero_of_xOnly_divFree`, line 185)**.** *If $a, b$ are XOnly and divergence-free then $[a,b] = 0$.*

*Proof.* For XOnly $a$, divergence-freeness reads $p_0(\hat{a}_p)_0 = 0$ for every $p$; similarly $q_0(\hat{b}_q)_0 = 0$. Fix $p, q$, $v = \hat{a}_p$, $w = \hat{b}_q$. Then $v \cdot q = q_0 v_0$ and $w \cdot p = p_0 w_0$, so $\mathrm{bvec}(p,q,v,w) = \mathrm{i}(q_0 v_0\, w - p_0 w_0\, v)$. Its components $1, 2$ vanish since $v_j = w_j = 0$ there; its component 0 is $\mathrm{i}(v_0\, q_0 w_0 - w_0\, p_0 v_0) = 0$ by the two divergence conditions (Lean: `linear_combination` $v_0 \cdot h_2 - w_0 \cdot h_1$). Hence every summand of $[a,b]$ is zero. □

**Lemma 11** (`mixedF_eq_zero_of_shear`, `shearY_mixed_cancel`; lines 207, 325)**.** *If $a, b, c$ are XOnly and divergence-free, then $S(a,b,c) = 0$ at $u = u_Y$, for all $\kappa, w \in \mathbb{C}$.*

*Proof.* $u_Y$ is XOnly and divergence-free (`xOnly_uY`, `divFree_uY`: its modes are $\pm k_y$ with $x$-directed coefficients, and $(\pm k_y) \cdot (*,0,0) = 0$). Both properties are preserved by scalar multiples. In $M(a,b,c) = [T(a,b),c] + T([a,b],c)$: $[a,b] = 0$ by Lemma 10 and $T(0,c) = 0$; and $T(a,b) = s\,u_Y$ is XOnly and divergence-free, so $[su_Y, c] = 0$ by Lemma 10. Thus $M(a,b,c) = 0$; the same holds for the cyclic permutations, so $S = 0$. □

**Theorem 12** (`shearY_jacobi_holds`, line 410)**.** *For all $\kappa, w \in \mathbb{C}$ and all $a, b, c \in \mathsf{Fld}$ that are XOnly and divergence-free, $J(a, b, c) = 0$ for the tethered bracket at $u = u_Y = (\sin y, 0, 0)$.*

*Proof.* By Lemma 9, $\langle u_Y, u_Y \rangle \neq 0$, so Lemma 8 gives $J(a, b, c) = S(a, b, c)$, which is 0 by Lemma 11. □

*Remark* 13 (What the theorem does and does not say). (i) XOnly plus divergence-free means $\hat{a}_k \neq 0$ only for $k_0 = 0$: these are the shear flows $f(y, z)\, e_x$. On this class the classical Lie bracket is zero. The tether $T(a, b) = s\, u_Y$ can be nonzero, but it lies in $\mathbb{C}u_Y$, inside the class, so every nested term ($[T(a, b), c]$, $T([a, b], c)$, $[[a, b], c]$, $T(T(a, b), c)$) vanishes individually. There is no cancellation *between* terms. This is an elementary checked boundary case, not a new Jacobi theorem. (ii) Theorem 14 below extends this to every XOnly, divergence-free $u$ with $\langle u, u \rangle \neq 0$. That condition is not automatic under the complex-bilinear pairing, and it cannot be dropped (Proposition 16). (iii) The class is *closed* under the operation: for $a, b$ in the class, $\{a, b\} = T(a, b) \in \mathbb{C}u_Y$, and $u_Y$ is XOnly and divergence-free. (iv) XOnly cannot be dropped: companion paper II gives an explicit divergence-free diagonal triple for which $J \neq 0$.

## 3 Every non-isotropic shear axis; non-vacuity; the isotropic guard

**Theorem 14** (`shear_jacobi_holds_of_pr_ne_zero`)**.** *Let $u \in \mathsf{Fld}$ be XOnly and divergence-free with $\langle u, u \rangle \neq 0$. For all $\kappa, w \in \mathbb{C}$ and all XOnly, divergence-free $a, b, c$, $J(a, b, c) = 0$.*

*Proof.* Lemma 8 applies because $\langle u, u \rangle \neq 0$, so $J = S$. The proof of Lemma 11 used only that $u$ is XOnly and divergence-free (Lean: `mixedSumF_eq_zero_of_shear`, which takes exactly these two hypotheses on $u$), so $S = 0$. Theorem 12 is the case $u = u_Y$ (Lemma 9). □

**Proposition 15** (non-vacuity; `brT_uY_eX_eX`, `paperI_nonvacuous`)**.** *Let $e_X = \delta_0(1, 0, 0)$, the constant field $(1, 0, 0)$. It is XOnly and divergence-free, and for all $\kappa, w$, $\{e_X, e_X\} = -(\kappa w)\, u_Y$, which is nonzero when $\kappa w \neq 0$.*

*Proof.* $e_X$ is supported at $k = 0$, so $k_{\mathbb{C}} \cdot \hat{e}_k = 0$ for all $k$. The modes of $u_Y$ are $\pm k_y \neq 0$, so $\langle u_Y, e_X \rangle = \sum_{p=\pm k_y} (\hat{u}_Y)_p \cdot (\hat{e}_X)_{-p} = 0$ and $\Pi_{u_Y} e_X = e_X$. Also $\langle e_X, e_X \rangle = 1$ and $[e_X, e_X] = 0$. Hence $\{e_X, e_X\} = 0 + T(e_X, e_X) = -(\kappa w) \cdot 1 \cdot u_Y$. Finally $u_Y \neq 0$ since its $k_y$ coefficient has first component $-\frac{\mathrm{i}}{2}$. □

So the hypothesis class of Theorems 12 and 14 is inhabited, and the theorem is not the trivial statement that every bracket in the class is zero: the tether term is nonzero there.

**Proposition 16** (the guard $\langle u, u \rangle \neq 0$ is needed; `shear_jacobi_fails_isotropic`)**.** *Let $u_{\mathrm{iso}} = \delta_{k_y}(1, 0, 0)$ (the complex field $e^{\mathrm{i}y} e_x$) and $c_Y = \delta_{k_y}(1, 0, 0) + \delta_{-k_y}(1, 0, 0)$ (the real field $(2 \cos y, 0, 0)$). Both are XOnly and divergence-free, $\langle u_{\mathrm{iso}}, u_{\mathrm{iso}} \rangle = 0$, and with the definitions above (including Lean's convention $x/0 = 0$),*

$$J(c_Y, c_Y, c_Y) = 6(\kappa w)^2\, u_{\mathrm{iso}},$$

*which is nonzero when $\kappa w \neq 0$.*

*Proof.* Divergence-freeness: the modes are $\pm k_y$ and $(0, \pm 1, 0) \cdot (1, 0, 0) = 0$. Since $-k_y \notin \operatorname{supp} u_{\mathrm{iso}}$, $\langle u_{\mathrm{iso}}, u_{\mathrm{iso}} \rangle = (1, 0, 0) \cdot (\hat{u}_{\mathrm{iso}})_{-k_y} = 0$; so $\Pi_{u_{\mathrm{iso}}} x = x - 0 \cdot u_{\mathrm{iso}} = x$ for all $x$ (division by zero is 0). Next $\langle c_Y, c_Y \rangle = 1 + 1 = 2$ and $\langle u_{\mathrm{iso}}, c_Y \rangle = (1, 0, 0) \cdot (\hat{c}_Y)_{-k_y} = 1$. By Lemma 10, $[c_Y, c_Y] = 0$, so $\{c_Y, c_Y\} = T(c_Y, c_Y) = -2\kappa w\, u_{\mathrm{iso}}$. Put $s = -2\kappa w$. Again by Lemma 10, $[s u_{\mathrm{iso}}, c_Y] = 0$, so $\{s u_{\mathrm{iso}}, c_Y\} = T(s u_{\mathrm{iso}}, c_Y) = -(\kappa w)\, s\, \langle u_{\mathrm{iso}}, c_Y \rangle\, u_{\mathrm{iso}} = 2(\kappa w)^2 u_{\mathrm{iso}}$. The three cyclic terms of $J(c_Y, c_Y, c_Y)$ are equal, so $J = 6(\kappa w)^2 u_{\mathrm{iso}}$; and $u_{\mathrm{iso}} \neq 0$. □

The nonzero value is exactly the tether–tether term $T(T(a,b),c)$, which Lemma 8 removes only when $\langle u,u\rangle \neq 0$. The statement is about the operation exactly as defined, including the division convention. It says nothing about any other choice of $\Pi_u$ for isotropic $u$, and $u_{\mathrm{iso}}$ is not a real field.

## 4 Lean mapping

| Statement | Lean name (`FieldJacobi.lean`:line, or Companion file) | Exact hypotheses |
|---|---|---|
| Thm. 12 | `shearY_jacobi_holds`:410 | `{κ w : ℂ} {a b c : Fld}`, `XOnly a`, `DivergenceFree a`, same for `b`, `c`; concludes `jacT κ w uY a b c = 0` |
| Lem. 11 | `shearY_mixed_cancel`:325; `mixedSumF_eq_zero_of_shear`:214 | as above; general version also takes `XOnly u`, `DivergenceFree u` |
| Lem. 10 | `br_eq_zero_of_xOnly_divFree`:185 | `XOnly`, `DivergenceFree` for both |
| Lem. 8 | `jacT_eq_mixedSum`:384 | `pr u u ≠ 0` |
| Lem. 9 | `pr_uY_uY`:396 | none |
| (classical Jacobi) | `jacF_eq_zero`:164 | none |
| Thm. 14 | `shear_jacobi_holds_of_pr_ne_zero` (Companion) | `XOnly u`, `DivergenceFree u`, `pr u u ≠ 0`, and `XOnly`, `DivergenceFree` for `a b c` |
| Prop. 15 | `brT_uY_eX_eX`, `brT_uY_eX_eX_ne_zero`, `paperI_nonvacuous` (Companion) | none; $\kappa w \neq 0$ for $\neq 0$ |
| Prop. 16 | `pr_uIso_uIso`, `jacT_uIso_cY`, `shear_jacobi_fails_isotropic` (Companion) | $\kappa w \neq 0$ |

## 5 Provenance, reproduction and axioms

- Full Lean source: `https://github.com/BenFrohman/frohmanian-tether-verified`, code commit `cf04afd5cd9b314703a5c8b196d83ec6b7979e6d` (pull request #2; Apache-2.0). The definitions and the original theorems are in `FTVerified/May/FieldJacobi.lean` and `JacobiWitness.lean`, unchanged since `26f16da`; the new theorems are in `FTVerified/Companion/JacobiCompanion.lean`. Line numbers refer to these files. Geometric comparison repository: `BenFrohman/frohmanian-tether-geometric` at `9e40ae7`.
- Lean toolchain (`lean-toolchain`): `leanprover/lean4:v4.30.0-rc1`.
- Mathlib (`lake-manifest.json`): input rev `v4.30.0-rc1`, locked rev `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`.

One-command rebuild (repository root, at the commit above):

```
lake exe cache get && lake build && python3 scripts/check_jacobi_axioms.py
```

GitHub Actions ran this, together with a comment-aware scan for `sorry`, `admit`, `native_decide` and `axiom` over all 31 project modules, in run `https://github.com/BenFrohman/frohmanian-tether-verified/actions/runs/42923330722` (passed: full library build of 8295 jobs; `PASS: 49 declarations, standard axioms only`). The axiom lines for the declarations used in this paper are (full names are the bracketed namespace prefix `Frohman.Tether.` plus the quoted name):

```
[Frohman.Tether.FieldJacobi]
  'FieldJacobi.jacF_eq_zero'
  'FieldJacobi.jacT_eq_mixedSum'
  'FieldJacobi.pr_uY_uY'
  'FieldJacobi.br_eq_zero_of_xOnly_divFree'
  'FieldJacobi.mixedSumF_eq_zero_of_shear'
  'FieldJacobi.shearY_mixed_cancel'
  'FieldJacobi.shearY_jacobi_holds'
[Frohman.Tether.JacobiCompanion]
  'JacobiCompanion.shear_jacobi_holds_of_pr_ne_zero'
  'JacobiCompanion.brT_uY_eX_eX'
  'JacobiCompanion.brT_uY_eX_eX_ne_zero'
  'JacobiCompanion.paperI_nonvacuous'
  'JacobiCompanion.pr_uIso_uIso'
  'JacobiCompanion.jacT_uIso_cY'
  'JacobiCompanion.shear_jacobi_fails_isotropic'
  each: depends on axioms: [propext, Classical.choice, Quot.sound]
```

The checker fails the build if any audited declaration depends on anything beyond `propext`, `Classical.choice` and `Quot.sound`.

## 6 Comparison with the geometric repository

The repository `frohmanian-tether-geometric` at commit `9e40ae7` formalizes a different object. Line numbers below refer to that commit.

- *Geometric bracket* (`SymplecticTether.lean` 353–381). It is a *scalar* bracket of functionals, $\{F, G\} = \text{ClassicalBracket}(F, G)(\omega) + \text{TetherKernel}(\omega, F, G)$, with

$$\text{TetherKernel}(\omega, F, G) = -\kappa\, \text{Enstrophy}(\omega) \int \left\langle \Pi_{u(\omega)}\, \delta F(x),\ \Pi_{u(\omega)}\, \delta G(x) \right\rangle dx.$$

  The tether axis is state-dependent: $u(\omega)$ is `velocity_from_vorticity` $\omega$, the Biot–Savart velocity (`ArnoldGeometric.lean` 918).

- *Parameters.* The geometric bracket has no free parameter $w$. Its tether prefactor is $-\kappa \cdot \text{Enstrophy}(\omega)$, with $\kappa$ fixed at $3/2$ (`kappa_eq_three_halves`, `SymplecticTether.lean` 300–310), and $\text{Enstrophy}(\omega) = \int \|\omega\|^2$ is a state functional (`ArnoldGeometric.lean` 208–213).

- *Projection.* $\Pi_u$ is the global $L^2$ Gram–Schmidt projector $v \mapsto v - \frac{\int v \cdot u}{\int \|u\|^2} u$ (`ForMathlib/Projection.lean` 35).

- *Functional derivative.* $\delta F$ is a chosen Gâteaux representative in the vorticity pairing, and it is 0 if no representative exists (`FunctionalDerivative`, `ArnoldGeometric.lean` 313). For the kinetic energy the representative is the stream function $\psi$, not the velocity (`SymplecticTether.lean` 366–368).

- *Fourier model (here).* $u$ is *frozen*, $\kappa, w \in \mathbb{C}$ are free, and $\{a, b\}$ is a *vector-valued* operation on velocity-slot fields.

**Shared features (stated, not an identification).** The geometric operation is also not antisymmetric, because its correction is symmetric. Its diagonal tether is negative semidefinite: $\{F, F\} = T(F, F) = -\kappa\|\omega\|^2 \int \|\Pi_u \delta F\|^2 \leq 0$ (`SymplecticTether.lean` 375–380). No geometric Jacobi witness is proved in that repository. Its `tethered_jacobi_identity` (`SymplecticTether.lean` 1187–1196) is conditional on the hypotheses `hident` (the Jacobiator equals an integrated cyclic Lie pairing) and `hvanish` (that integral vanishes); its proof is the rewrite `rw [hident, hvanish]`.

**Why nothing transports.** Pairing the vector operation here with a state $m$ produces a factor $\langle m, u\rangle$ from the tether term. Even formally, the parameters could therefore be matched only pointwise, where that pairing is nonzero. No identification between the two models is claimed. *Informal remark (not proved):* because $u$, Enstrophy and $\Pi_{u(\omega)}$ depend on the state, one expects a geometric Jacobiator to contain extra terms involving $\delta E$ and $\delta\Pi$, which the frozen-$u$ `jacT` lacks. Neither the differentiability of the state-dependent $\Pi$ nor any geometric Jacobiator formula is proved in either repository. Neither the restricted identity (paper I) nor the counterexample (paper II) is claimed to transport to the geometric setting. This comparison is not a refutation, or a confirmation, of any geometric statement.

## 7 Audit scope

- The result concerns the *finite-Fourier field model* Fld of complex trigonometric-polynomial vector fields on $\mathbb{T}^3$, with the vector-valued bracket $\{a, b\} = [a, b] + T(a, b)$ defined above. It is not a statement about a Poisson bracket on functionals, about smooth or Sobolev fields, or about Navier–Stokes solutions.
- No identification of this model, or of the shear class, with the geometric reduced coadjoint orbit (or with $u^\perp$ in any geometric sense) has been proved in Lean or here. The link from Fourier coefficients to the `fderiv`-based fields of `Torus.lean` is not formalized for `br`.
- Kernel checking establishes that the stated Lean propositions follow from the stated definitions. It does not establish that the definitions faithfully model the intended geometry, nor any novelty of the results.
- Nothing here bears on any Clay Millennium Problem, and no such claim is made.

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## 8 Relation to companion paper II

Paper II (*Failure of Antisymmetry and of the Cyclic Jacobi Identity for a Specified Finite-Fourier Tethered Operation*) proves `shearY_jacobi_fails_transverse`: at the same $u_Y$, $J(a_Z, a_Z, a_Z) \neq 0$ whenever $\kappa w \neq 0$, with $a_Z = (0, 0, \sin x)$ divergence-free but not XOnly. The XOnly hypothesis in Theorem 12 therefore cannot be dropped. Explicitly, XOnly cannot be dropped, because an explicit divergence-free diagonal triple fails. No claim is made about divergence-free triples of three distinct arguments.